Fuel Booster Pump – Maintenance Practices

1. Removal/Installation

A. Remove Booster Pump

1) Ensure battery and/or electrical power supply is disconnected and crossfeed valve closed. Ensure WING TANK FUEL switches (if fitted) are OFF.

2) Drain applicable fuel tank (refer to 12-10-10, Fuel System – General Data).

3) Remove access panel from underside of fuselage (refer to 12-00-00, Servicing – General Data).

4) Disconnect pressure switch line from booster pump base and loosen line to facilitate pump removal.

WARNING **ON AIRCRAFT FITTED WITH E.M.I. FILTER, BEFORE REMOVING BOOSTER PUMP ELECTRICAL CONNECTIONS, LOAD TERMINAL ON MAIN DISTRIBUTION BOX OF ASSOCIATED INTERFERENCE FILTER MUST BE DISCHARGED TO AIRCRAFT GROUND.**

5) Before disconnecting electrical leads from terminal on booster pump, proceed as follows:

a) Pull out associated booster pump circuit breaker.

b) Discharge filter through LOAD terminal either by connecting LOAD terminal to aircraft ground through a 150 ohm, 7.5 watt resistor, for at least 30 seconds, or, connecting a wire to an aircraft ground and discharging by touching other end of wire to LOAD terminal.

Note Do not reengage circuit breaker until leads are reconnected to booster pump terminals.

If leads have to be subsequently removed for further maintenance, discharge procedures in Para 1.A.5) above must be repeated prior to lead disconnection.

6) Disconnect leads from, radio noise filter at pump base, disconnect leads from pressure switch. For Mod 6/2299 fuel pumps, disconnect the electrical fly leads from the quick disconnect splice ends.

7) Remove booster pump retaining bolts and washers, and lower pump until upper end of check valve is exposed.

8) Disconnect flexible line at check valve.

9) Remove booster pump and attached parts.

10) Remove check valve and seal drain tube from booster pump.

11) Cap all openings.

B. Install Booster Pump

1) Install check valve and seal drain tube to booster pump.

2) Position booster pump and gasket below sump plate so that radio noise filter at pump base is parallel to aircraft center line.

3) Connect flexible line to check valve.

4) Lift booster pump into position and install retaining bolts and washers.

5) Connect leads to radio noise filter and connect leads to pressure switch. For Mod 6/2299 fuel pumps, connect the electrical fly leads contacts to the individual quick disconnect splice ends.

   Note If leads have to be subsequently removed for further maintenance, discharge procedures in Para 1.A.5) must be repeated prior to lead disconnection.

6) If removed, connect pressure switch line to booster pump and tighten all line connections.

7) Check for fuel leaks (refer to 28-00-00, Fuel (Mod 6/2006) – General Data).

8) Install access panel (refer to 12-00-00, Servicing – General Data).

2. Adjustment/Test

A. Operational Test

1) Ensure each fuselage tank contains a minimum of 30 Imperial (36 U.S.) gallons of fuel.

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Ensure relevant circuit breakers are engaged.

4) Ensure fuel levers are OFF, FUEL SELECTOR switch is set to NORMAL, and wing tank fuel switches (if fitted) are OFF.

5) Set forward and aft BOOST PUMP 1 switches to ON. Check all No. 1 and No. 2 boost pump pressure caution messages go out and verify that the fuel pump amber blocks Pmp1 and Pmp2 for the forward and aft fuel tanks are not displayed.

   Note Fuel pressure from the No. 1 booster pump in each tank actuates a pressure switch which renders the Pmp2 amber blocks inoperative.

6) Hold forward and aft BOOST PUMP 1 switches in the TEST position. Check both No. 2 boost pump pressure amber blocks and caution messages disappear.

   Note Both No. 2 booster pumps are operating to actuate the No. 2 boost pump amber blocks Pmp2.

7) Release forward and aft BOOST PUMP 1 switches. Check all No. 1 and No. 2 boost pump amber blocks Pmp1 and Pmp2 and associated CAS messages are displayed.

8) With container in position to receive spillage, disconnect right engine fuel supply line at firewall connection.

9) Connect a suitable length of hose to firewall connection and feed into container.

10) Select BOOST PUMP 1 switch ON.

11) When fuel flow is stabilized through hose at engine nacelle, measure flow rate. (Minimum of 1 Imperial gallon in 30 seconds).

12) Hold BOOST PUMP 1 switch in TEST position and repeat Para 2.A.11).

13) Release BOOST PUMP 1 switch, remove hose from firewall connection and reconnect fuel line.

14) Repeat Para 2.A.8) through Para 2.A.13) for aft tank substituting AFT for FWD and left engine for right engine.

15) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Bench Test

Equipment required (As shown in Figure 1):

- Multi-meter
- 27 Volt DC Supply

Note Calibration is done under ambient conditions, using JP–4 Turbine Fuel as the test fluid and Figure 2 to check limits of results.

1) With test setup as shown in Figure 1 and temperature of fluid at 80° ± 10°F, run pump to 460 to 560 pph flow rate for a period of 15 minutes. Ensure max current limit (Figure 2) is not exceeded.

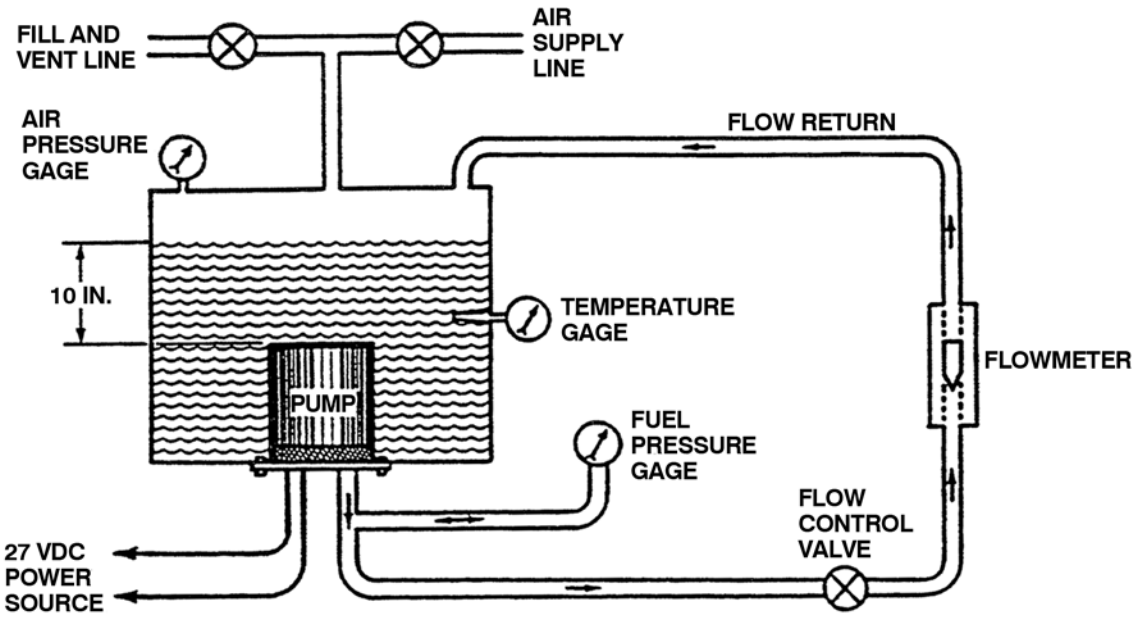
2) Check no leakage occurs through discharge port, tank drain port casting, gaskets, or around motor leads, during Para 2.B.1). A maximum leakage allowance of 5 drops (1/4 cc) in 1 hour is allowed for seal drain port while pump is running.

3) Close flow control valve and check fuel pressure does not exceed 40 psi.

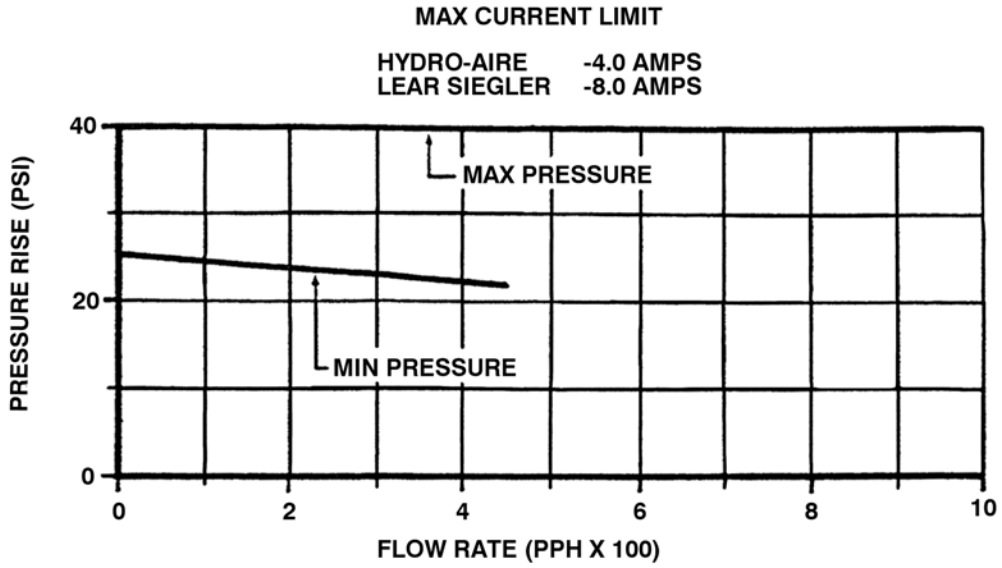
4) Open flow control valve to rate fuel flow at 450 pph check fuel pressure does not fall below 22 psi.

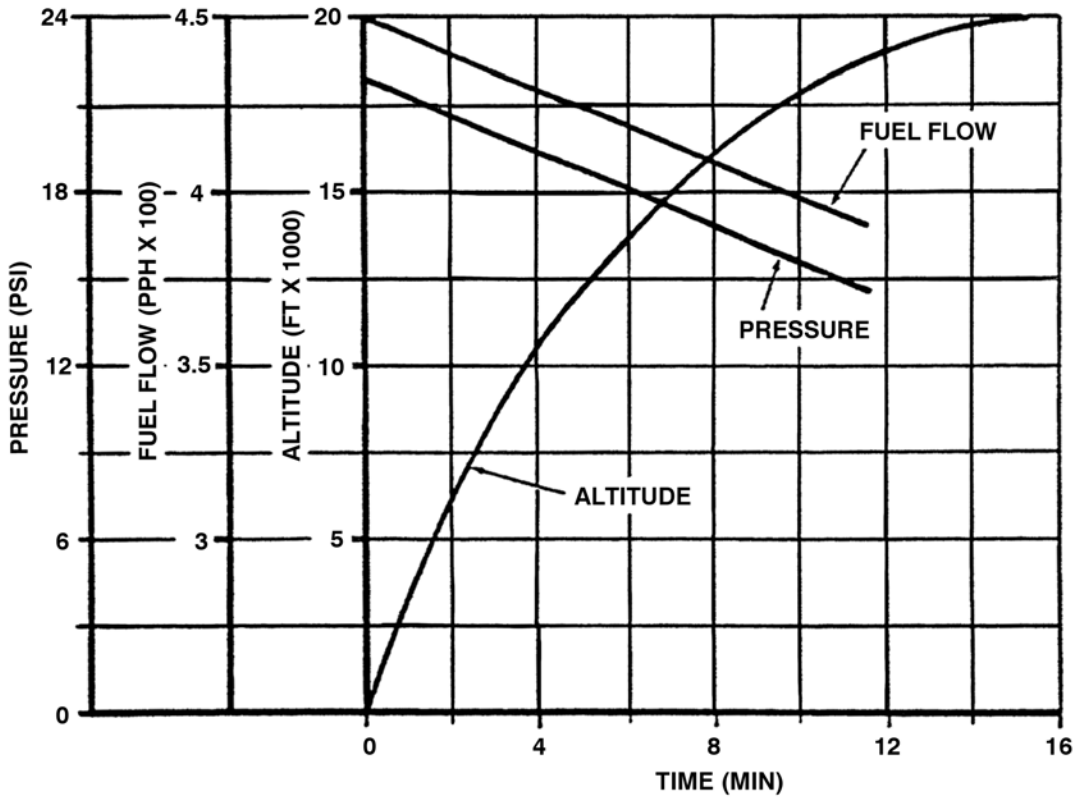
5) Switch off pump.

6) Apply 15 psi air pressure within tank for a period of 5 minutes and check no fuel leaks occur at areas listed in Para 2.B.2).



Pump Test Rig
Figure 1





Bench Test Curves
Figure 2

3. Inspection/Check

A. Leak Check – Seal Drain Port

1) Remove access panel from under side of fuselage (refer to 12-00-00, Servicing – General Data).

2) Ensure each fuselage tank contains a minimum of 30 Imperial (36 U.S.) gallons of fuel.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Ensure relevant circuit breakers are engaged.

5) Ensure fuel levers are OFF, FUEL SELECTOR switch is set to NORMAL, and WING TANK FUEL switches (if fitted) are OFF.

6) Select forward and aft BOOST PUMP 1 switches to ON. Check all No. 1 and No. 2 boost pump amber blocks Pmp1 and Pmp2 and associated CAS messages are not displayed.

7) Check each No. 1 pump seal drain port for fuel leaks. Maximum leakage allowed 5 drops (1/4 cc) in 1 hour while pump is running. No leakage allowed on stationary pump.

8) Switch off No. 1 pumps by selecting booster pump switches to OFF.

9) Select forward and aft BOOST PUMP 2 switches to EMERGENCY. Check both No. 2 boost pump amber blocks Pmp2 are no longer displayed.

10) Repeat Para 3.A.7) for the No. 2 pumps.

11) Switch forward and aft BOOST PUMP 2 switches to OFF.

12) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

13) Install access panel to underside of fuselage (refer to 12-00-00, Servicing – General Data).

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